

Thomas Ford

Senior AI Engineer

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SUMMARY

Senior AI Engineer with deep experience building and deploying Generative AI, LLM, and RAG solutions across healthcare operations, patient engagement, and clinical workflows. Skilled in Python, prompt engineering, agentic frameworks, GCP, and MLOps practices including MLFlow, Docker, and CI/CD pipelines. Introduced 6 LLM prototypes at CVS Health that cut investigation cycles by roughly 30%, and reduced patient-service processing time by 24% through AI-augmented API redesign. Brings a responsible-AI mindset and cross-functional track record translating complex business requirements into scalable, auditable GenAI systems.

SKILLS

Backend: Python, Node.js, REST APIs, T-SQL, PySpark, Express.js, Event-Driven Architecture, LLM Orchestration

Frontend: React, TypeScript, Redux, Material UI, HTML5, CSS3, Accessible Forms, Responsive Web Applications

Cloud: GCP, Vertex AI, AWS, Docker, Kubernetes, GitHub Actions, CI/CD Pipelines, CloudWatch

Data: Snowflake, PostgreSQL, SQL Server, Redis, MLFlow, Data Modeling, SQL Query Optimization, Audit Logging

Tools: MLOps, Prompt Engineering, RAG, Agentic Frameworks, GitLab, Postman, Swagger/OpenAPI, Jest

Industry: Generative AI, LLM Security, Responsible AI, Value-Based Care, Revenue Cycle Management, HIPAA Compliance, EHR Integration, Clinical Workflow Automation

WORK HISTORY

Senior AI Engineer

CVS Health

05/2023 - Present

Key Qualifications & Responsibilities

- Architected and deployed 6 production-ready Generative AI solutions using Python, Vertex AI, and GPT-based LLMs across 12 patient-facing workflows spanning scheduling, virtual care, prescription summarization, and benefits eligibility, collaborating with product, clinical, and compliance stakeholders.
- Designed end-to-end RAG pipelines and agentic frameworks integrated with PostgreSQL, Redis, and REST APIs to automate complex care-journey workflows for MinuteClinic® and CVS Virtual Care, reducing manual outreach volume by 18% across operations teams.
- Implemented MLOps best practices across all GenAI deployments including MLFlow model versioning, drift analysis, Docker containerization, and GitHub Actions CI/CD pipelines, supporting 4 concurrent engineering workstreams on GCP and AWS infrastructure.

Key Achievements

- Increased appointment-flow completion by 31% by engineering LLM-powered registration guidance with prompt engineering and retrieval-augmented context injection across CVS Virtual Care patient journeys.
- Reduced patient-service processing time by 24% by deploying Python-based agentic workflows that unified scheduling, prescription, and benefits data retrieval behind a single orchestrated GenAI API layer with Snowflake-backed caching.
- Introduced 6 LLM prototypes for support summarization, automated Jest test generation, and workflow-gap analysis, cutting selected investigation cycles by roughly 30% for clinical operations and engineering teams.
- Stabilized GenAI-augmented care journeys at 99.94% availability by redesigning LLM output moderation, access controls, and CloudWatch alerting across containerized Vertex AI services.

- Accelerated model iteration cycles by 41% by standardizing prompt versioning, quantization workflows, and responsible-AI guardrails including output moderation and auditability controls aligned with HIPAA and PHI handling requirements.

Skills: *Python, Vertex AI, RAG, Prompt Engineering, MLFlow, Docker, GCP, LLM Security*

Senior Software Engineer

BRIDGE Hospice

02/2021 - 05/2023

Key Qualifications & Responsibilities

- Led architecture for 9 patient-intake and care-coordination workflows using Python, React, C#/.NET, and SQL Server, delivering AI-augmented document summarization and clinical task-routing features in close coordination with nurses, physicians, intake coordinators, and billing stakeholders.
- Unified 5 legacy EHR, document, and notification integrations behind secure REST API contracts and introduced LLM-assisted digitization prototypes that reduced manual data-entry touchpoints by roughly 35% across the patient-admission pipeline.
- Established CI/CD release ownership for 3 production applications through Jenkins pipelines, Swagger/OpenAPI contracts, role-based access controls, audit logging, and cross-functional incident reviews aligned with HIPAA and Responsible AI governance requirements.

Key Achievements

- Streamlined referral validation and intelligent task routing by deploying a Python-based classification model, cutting intake handoff time in half while preserving nurse review checkpoints and PHI-aligned audit controls.
- Modernized 14 React screens and backend endpoints with AI-assisted form pre-population, improving first-pass intake completion by 27% across clinical and administrative user cohorts.
- Reduced peak SQL Server response time by 38% through query restructuring, index tuning, and background-job scheduling optimized for care-coordination dashboard data retrieval.
- Eliminated 42% of repeat production defects by expanding xUnit, Postman integration, and Cypress test coverage across admission, document-processing, and provider-communication paths.
- Resolved one-third of admission-status escalations by exposing LLM-generated workflow-state summaries and operational alerts to coordinators, clinicians, and product owners through a unified REST API layer.

Skills: *Python, React, C#/.NET, SQL Server, REST APIs, Jenkins, EHR Integration, HIPAA Compliance*

Software Engineer

UMC Physicians

07/2018 - 12/2020

Key Qualifications & Responsibilities

- Designed and delivered 8 MyTeamCare patient-portal capabilities across online scheduling, paperless registration, provider messaging, and medical-record access using JavaScript, React, REST APIs, and SQL, engaging physicians, front-desk staff, analysts, and infrastructure teams across 16 EHR data-mapping integrations.
- Standardized 2 portal release trains through Git, Jira, API documentation, automated regression tests, and Linux deployment runbooks, achieving a release-defect reduction of 29% over a 12-month delivery cycle.
- Initiated AI-assisted search and recommendation prototypes using Python scripting and SQL-backed data pipelines to accelerate provider-directory lookups to sub-second median response time for patient and staff users.

Key Achievements

- Transformed appointment booking into a single-session guided flow backed by eligibility-aware business logic, raising successful self-service scheduling by 33% without increasing call-center staffing.
- Simplified paperless registration forms through data-driven field validation and conditional rendering, reducing incomplete submissions by 22% while improving mobile accessibility scores.
- Implemented secure session management, authorization controls, and input validation that lowered account-access incidents by approximately 20% across patient and staff user populations.
- Revitalized 11 legacy portal modules using reusable React components and integration-test coverage, decreasing release-related regressions by 29% over two consecutive release cycles.

- Generated 13 operational and utilization reports backed by optimized SQL queries and targeted caching, shortening weekly data-reconciliation cycles by roughly 35% for analytics and operations stakeholders.

Skills: *Python, JavaScript, React, REST APIs, SQL, HTML5, Linux, Patient Engagement*

Junior Software Engineer

GermBlast

07/2016 - 06/2018

Key Qualifications & Responsibilities

- Developed 10 scheduling and field-operations modules using JavaScript, C#/.NET, MySQL, and responsive web forms for dispatchers, technicians, and customer-service staff, enforcing 15 validation and permission rules across customer, appointment, and service-record workflows.
- Launched automated email notification pipelines covering every appointment lifecycle state, reducing missed service visits by 17% while coordinating acceptance criteria directly with operations and field-team stakeholders.
- Initiated a biweekly release cadence for internal web applications through Git-based code reviews, unit-test coverage expansion, deployment checklists, and structured feedback sessions with operations users across 19 role-based permission profiles.

Key Achievements

- Improved dashboard load time by 41% by rewriting MySQL queries and eliminating unnecessary data retrieval across service-location views, directly reducing dispatcher response latency during peak scheduling windows.
- Solved recurring scheduling conflicts by implementing database constraints and deterministic assignment rules, lowering duplicate bookings by 26% over a 6-month measurement period.
- Expanded role-based permissions across 19 user profiles, cutting unauthorized record-change incidents to zero during the final year of the engagement.
- Generated 13 operational reports for invoicing, service history, and technician utilization that shortened weekly reconciliation cycles by roughly 35% for operations and finance stakeholders.
- Restructured data validation logic across customer and appointment workflows, reducing form-submission error rates by 19% and cutting operations team correction requests by more than one-third.

Skills: *JavaScript, C#/.NET, MySQL, HTML5, CSS3, Git, Unit Testing, REST APIs*

EDUCATION

Bachelor of Science in Computer Science

University of Kansas

08/2012 - 05/2016 | Lawrence, KS